

DOSSO, Niger

WMO: 610530

Lat: 13.049N

Lon: 3.223E

Elev: 225

StdP: 98.65

Time Zone: 1.00 (E01)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.6	17.7	-7.9	2.0	29.5	-7.2	2.1	28.7	6.3	24.3	5.7	25.0	1.3	N/A	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	12.6	41.9	21.9	41.1	21.7	40.2	21.7	27.1	34.3	26.7	33.6	26.4	33.1	2.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.6	21.5	29.2	25.2	20.9	28.8	24.9	20.5	28.7	86.7	34.3	85.0	33.7	83.4	33.0	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.4	4.6	4.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			6.9	28.3	0.7	1.3	6.4	29.2	6.0	29.9	5.6	30.6	5.1	31.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.6	25.1	28.5	32.2	34.6	34.1	31.5	28.8	27.5	28.7	30.5	28.7	25.5
	DBStd	3.48	2.18	2.41	1.80	1.60	2.12	2.25	1.82	1.66	1.69	1.40	1.22	1.90
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	7168	467	519	688	739	747	645	583	541	561	636	562	480
	CDD18.3	4127	208	286	430	489	489	395	325	283	311	377	312	222
	CDH23.3	55163	2038	3560	6434	7947	7771	5741	3920	2882	3618	5036	3966	2251
	CDH26.7	32366	935	2017	4199	5576	5319	3447	1799	1056	1650	2878	2419	1071
Wind	WSAvg	1.8	2.1	2.1	1.8	2.0	2.4	2.3	1.9	1.5	1.3	1.4	1.3	2.0
Precipitation	PrecAvg	523	0	0	2	4	30	77	146	172	79	12	0	0
	PrecMax	739	0	0	15	18	121	151	239	405	176	35	3	0
	PrecMin	355	0	0	0	0	1	26	61	104	15	0	0	0
	PrecStd	109	0	0	3	5	31	34	52	76	44	12	1	0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.7	40.1	41.9	43.0	42.8	40.2	37.2	34.7	36.6	39.0	38.8	36.9
		MCWB	17.4	19.4	19.9	21.8	23.7	25.3	25.7	26.1	26.1	22.1	19.8	18.0
	2%	DB	34.9	38.6	41.0	42.2	41.5	38.7	35.5	33.4	35.3	38.0	38.1	35.1
		MCWB	16.2	18.4	19.6	21.7	23.9	25.4	25.3	25.9	26.2	22.6	19.6	17.1
	5%	DB	33.3	37.1	40.2	41.5	40.5	37.6	34.2	32.4	34.3	37.2	37.3	33.8
		MCWB	15.3	17.5	19.4	21.5	24.3	25.4	25.2	25.7	26.1	23.0	19.4	16.4
	10%	DB	31.6	35.5	39.2	40.6	39.4	36.5	33.0	31.3	33.2	36.2	36.3	32.1
		MCWB	14.5	16.9	18.8	21.3	24.6	25.2	25.2	25.4	25.9	23.6	18.8	15.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.2	23.5	24.6	26.0	27.5	27.4	26.9	27.0	27.7	27.3	24.0	19.0
		MCDB	34.5	32.5	34.8	38.7	38.4	35.8	33.4	32.4	33.4	34.1	33.9	34.6
	2%	WB	16.9	22.0	23.6	25.2	26.7	26.7	26.2	26.5	27.0	26.6	22.5	18.0
		MCDB	33.8	32.3	35.2	37.8	36.8	34.7	32.0	31.4	32.5	33.6	33.3	34.0
	5%	WB	15.7	19.7	22.1	24.6	26.2	26.2	25.9	26.1	26.6	26.1	21.4	17.0
		MCDB	32.0	33.3	35.7	36.8	36.1	34.1	31.5	30.7	32.0	33.0	32.5	32.5
	10%	WB	14.7	17.7	19.7	24.1	25.8	25.9	25.6	25.8	26.2	25.7	20.1	16.1
		MCDB	30.5	34.6	37.5	35.8	35.5	33.7	31.0	30.0	31.5	32.3	32.5	30.8
Mean Daily Temperature Range	5% DB	MDBR	13.7	14.1	14.5	12.6	10.7	9.2	8.2	7.3	8.9	12.4	16.5	14.0
		MCDBR	15.9	15.9	15.8	13.8	11.8	10.4	9.7	8.8	10.3	13.8	17.4	16.2
		MCWBR	6.3	5.8	5.7	4.8	3.2	2.5	2.4	2.7	2.9	3.5	4.8	5.9
	5% WB	MCDWR	15.9	13.7	13.9	11.5	10.6	9.5	8.6	8.1	9.7	11.2	16.1	16.0
		MCWBR	6.6	5.9	6.8	4.2	3.3	2.8	2.7	3.1	3.3	3.2	5.7	6.1
Clear-Sky Solar Irradiance	taub	0.606	0.662	0.739	0.742	0.710	0.671	0.605	0.579	0.576	0.622	0.551	0.541	
	taud	1.686	1.539	1.418	1.495	1.553	1.640	1.826	1.876	1.872	1.729	1.773	1.789	
	Ebn at Noon	704	685	648	645	653	671	720	749	754	706	742	743	
	Edh at Noon	238	286	329	304	281	255	212	205	206	233	216	210	
All-Sky Solar Radiation	RadAvg	5.51	5.94	6.25	6.41	6.25	5.98	5.73	5.42	5.91	6.05	5.75	5.42	
	RadStd	0.25	0.26	0.33	0.23	0.18	0.21	0.18	0.24	0.21	0.15	0.14	0.21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	+0.37	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page